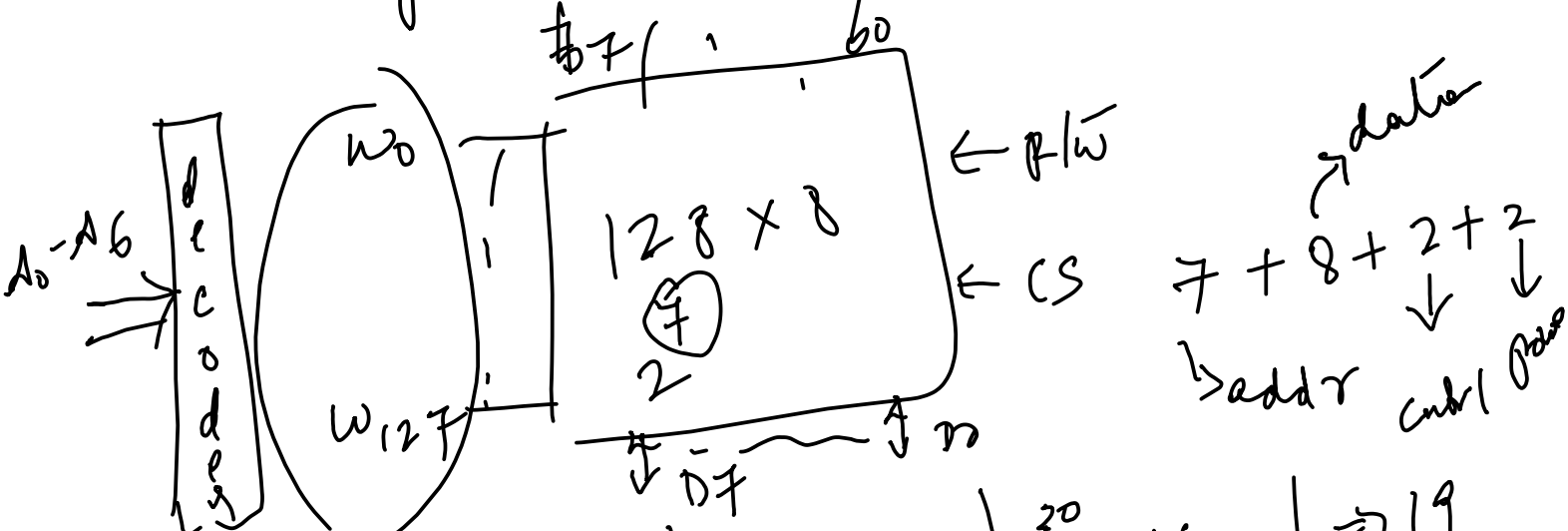
128 × 8 — memory

no. of external connections ?



$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1k$

$2^{12} = 4k$

$2^{17} = 128k$

$2^{20} = 1M$

$2^{22} = 4M$

$2^{27} = 128M$

$2^{30} = 1G$

$2^{32} = 4G$

$2^{37} = 128G$

$\Rightarrow 19$

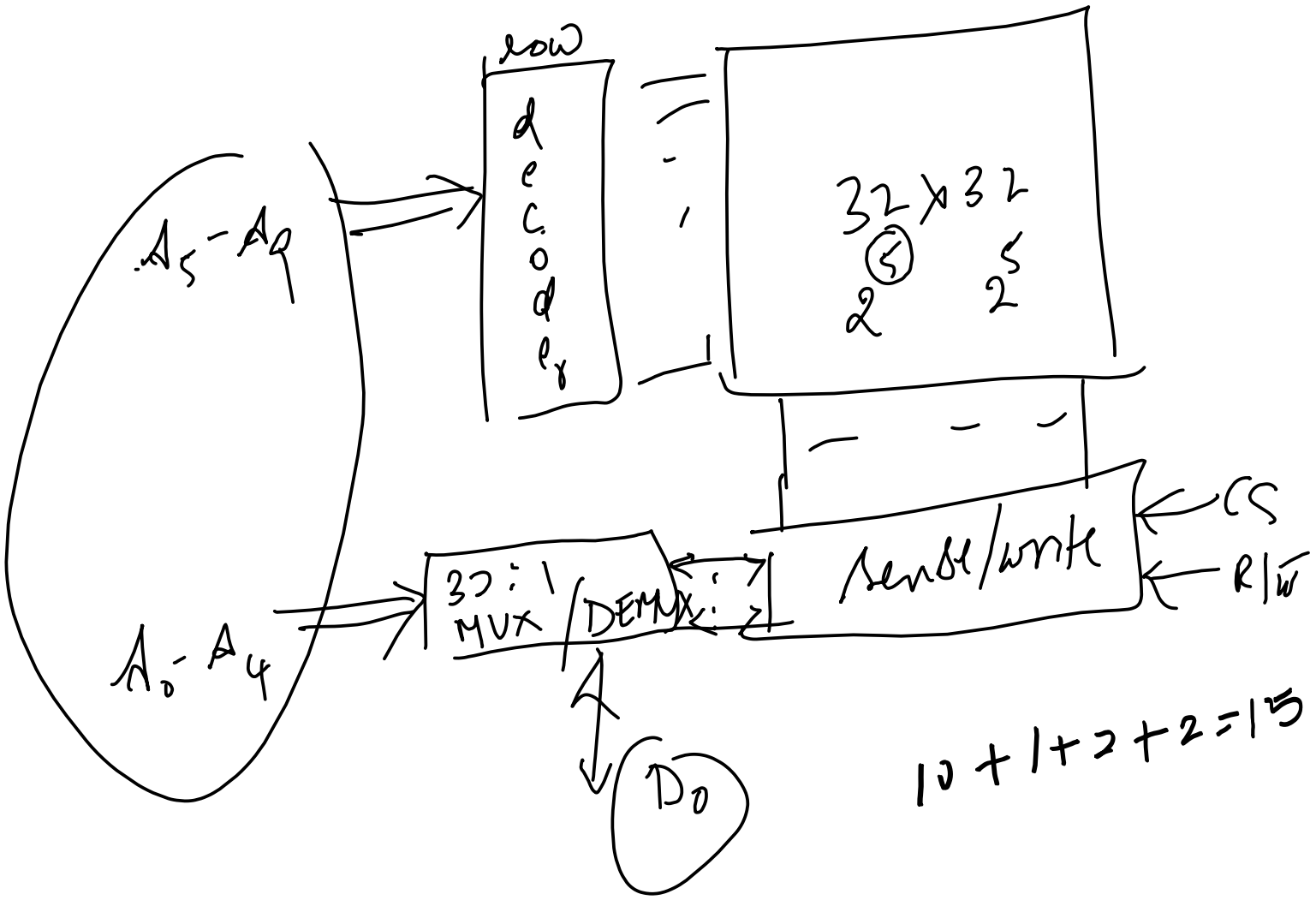


Mem organization

$32 \times 32 = 1024$ bits

$= 1$ kbit

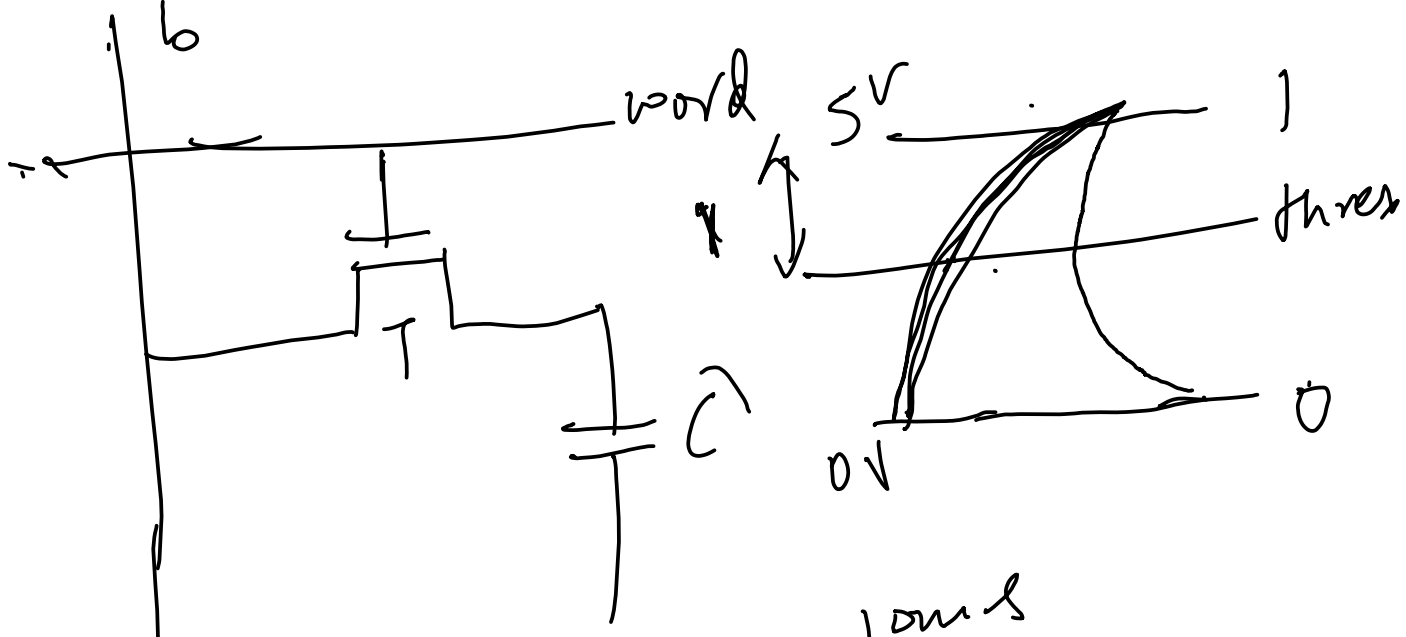
$A_0 - A_9$

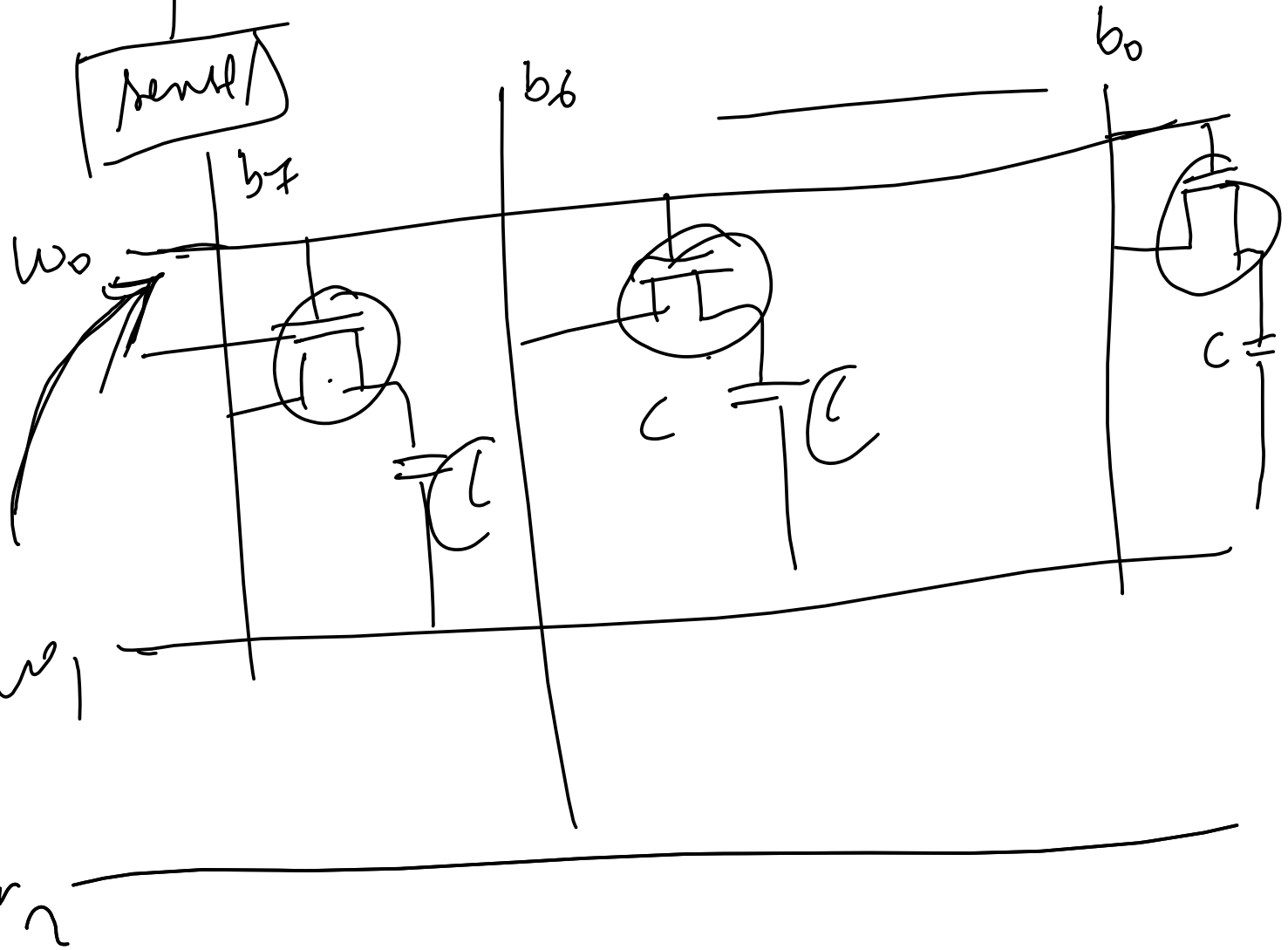


$10 + 1 + 2 + 2 = 15$

DRAM

Read





Write the internal organization of 256M bit DRAM chip (32M×8)

↳ Total capacity

32M×8

32M×8

$2^{25} \times 8$ Address line = 25

$= A_0 - A_{24}$

256M

16k

14

2

Row $(A_{11} - A_{24})$

$\frac{16k}{8} = \frac{2^{14}}{2^3} = 2^{14-3} = 2^{11}$

$16^2 = 256$

col $-(A_0 - A_{10})$

11

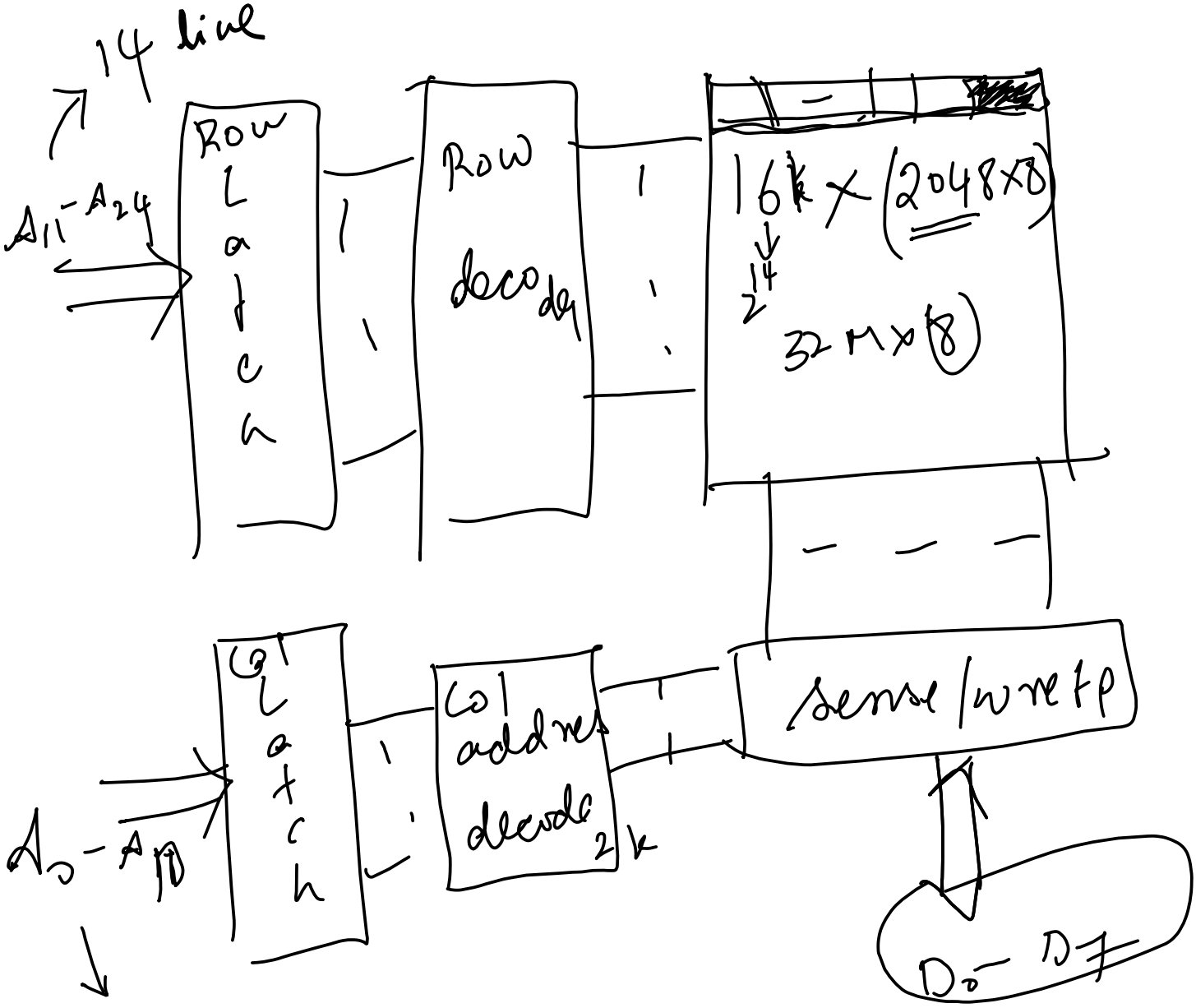
$= 2048$ bytes

2047

byte byte

16k × (2048×8)

16k × 16k


14 line
Row latch
Row decoder
16k × (2048×8)
2^{14}
32M×8
A_{11} - A_{24}
Col latch
Col address decoder
2k
Sense/write p
A_0 - A_{10}
11 lines
D_0 - D_7

$2^{11} = 2048 =$

Write the internal organization of 16M bits DRAM chip (2M × 8).

$2^{21} \rightarrow (A_0 - A_{20})$

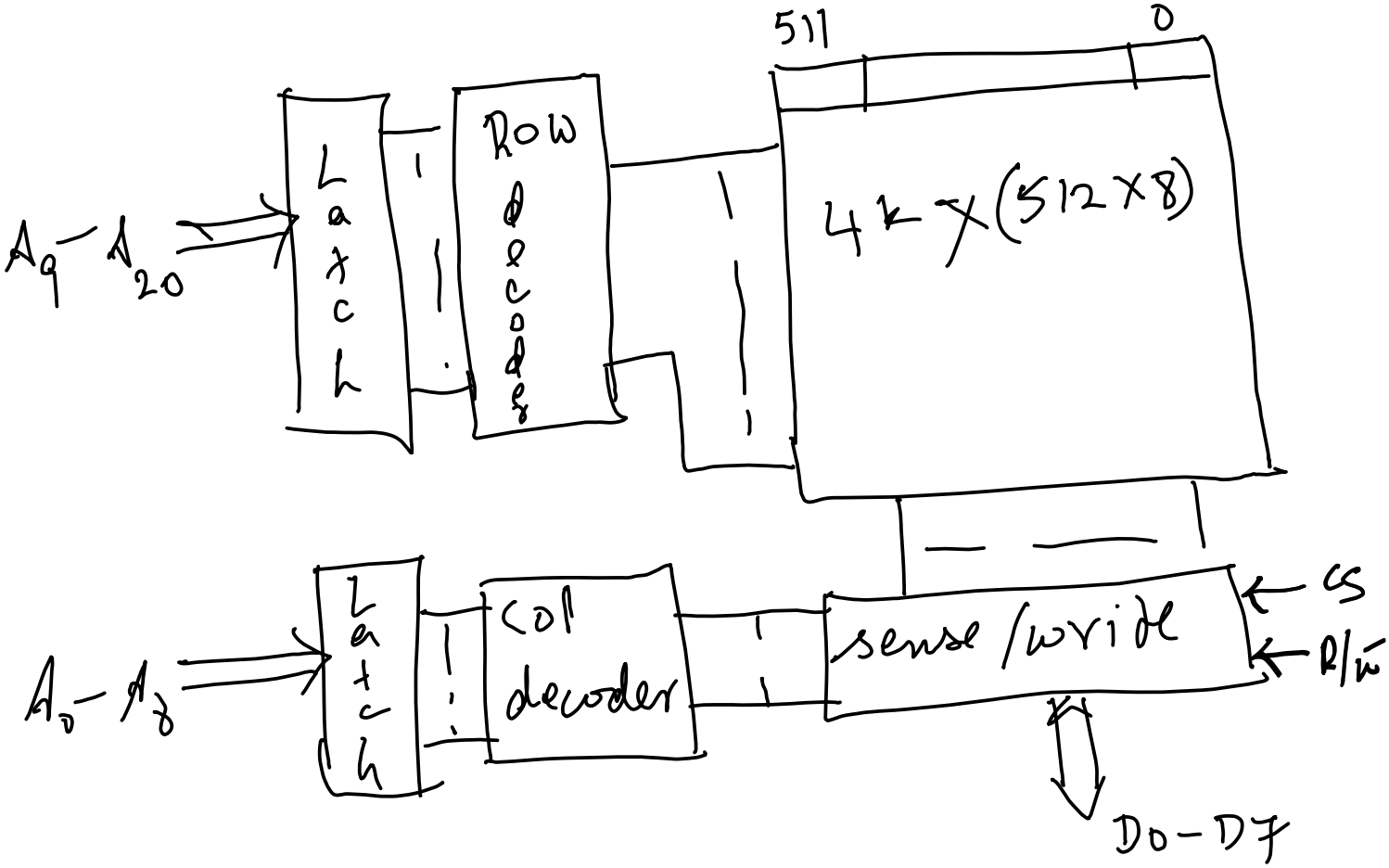
16M

4K

$\frac{4K}{8} = \frac{2^{12}}{2^3} = 2^{12-3} = 2^9$ (9) → col → $A_0 - A_8$

4K

(12) → row → $A_9 - A_{20}$



$21 + 8 + 2 + 2 = 33$ external connections

* 16 Mbit — (2M) × 8 $\Rightarrow 2^{21} \times 2^{3} = 2^{24}$

$\Rightarrow 2^{12} \times \frac{2^{12}}{2^{3}}$

(21) $\rightarrow A_0 - A_{20}$

$2^0 = 1$

$2^1 = 2, 2^{11}, 2^{21}, 2^{2M}$

$\Rightarrow 2^{12} \times 2^{9}$

Row $\rightarrow A_9 - A_{20}$ $A_0 - A_8$

Col

$A_9 - A_{20}$

$2^{12} \times 2^{9}$ bytes

4096×512 bytes

$A_0 - A_7$

$32M \times 8 \Rightarrow 2^{25} \times 2^{3} = 2^{28} = 2^{14} \times 2^{14}$

2^{3}

$\rightarrow A_0 - A_{24}$

$A_{11} - A_{24} \Rightarrow 2^{14} \times 2^{11} \rightarrow A_0 - A_{10}$

(row) (col)

$2^{14} \times 2^{11}$ bytes

16K X 2K bytes

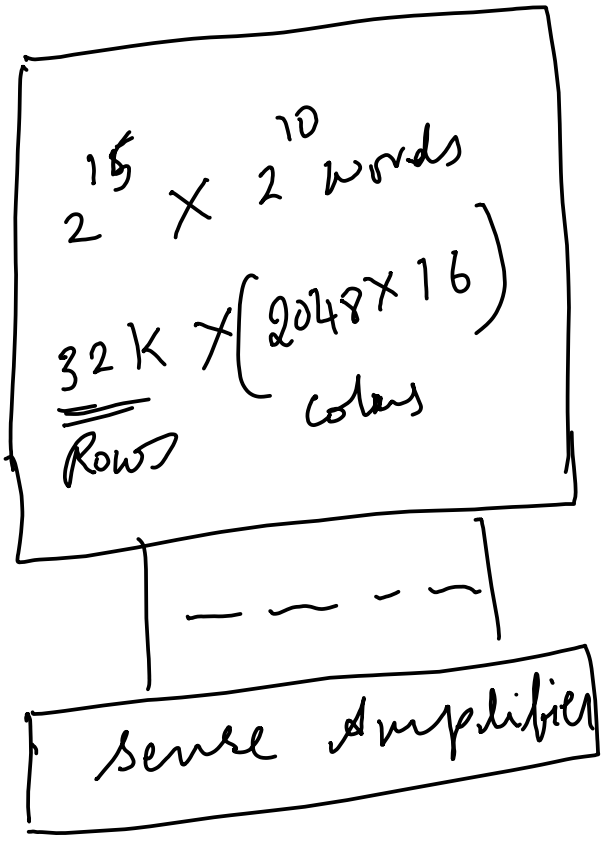
32M X 16 $\Rightarrow$ $2^{25} \times 2^{4} = 2^{29} = 2^{15} \times 2^{14}$ / 16

25 $\rightarrow$ $A_0 - A_{24}$

$\Rightarrow 2^{15} \times \frac{2^{14}}{2^4} = 2^{15} \times 2^{10}$

15 $\rightarrow$ $A_{10} - A_{24}$ (row)

10 $\rightarrow$ $A_0 - A_9$ (col)



row

$A_{10} - A_{24}$ $\Rightarrow$

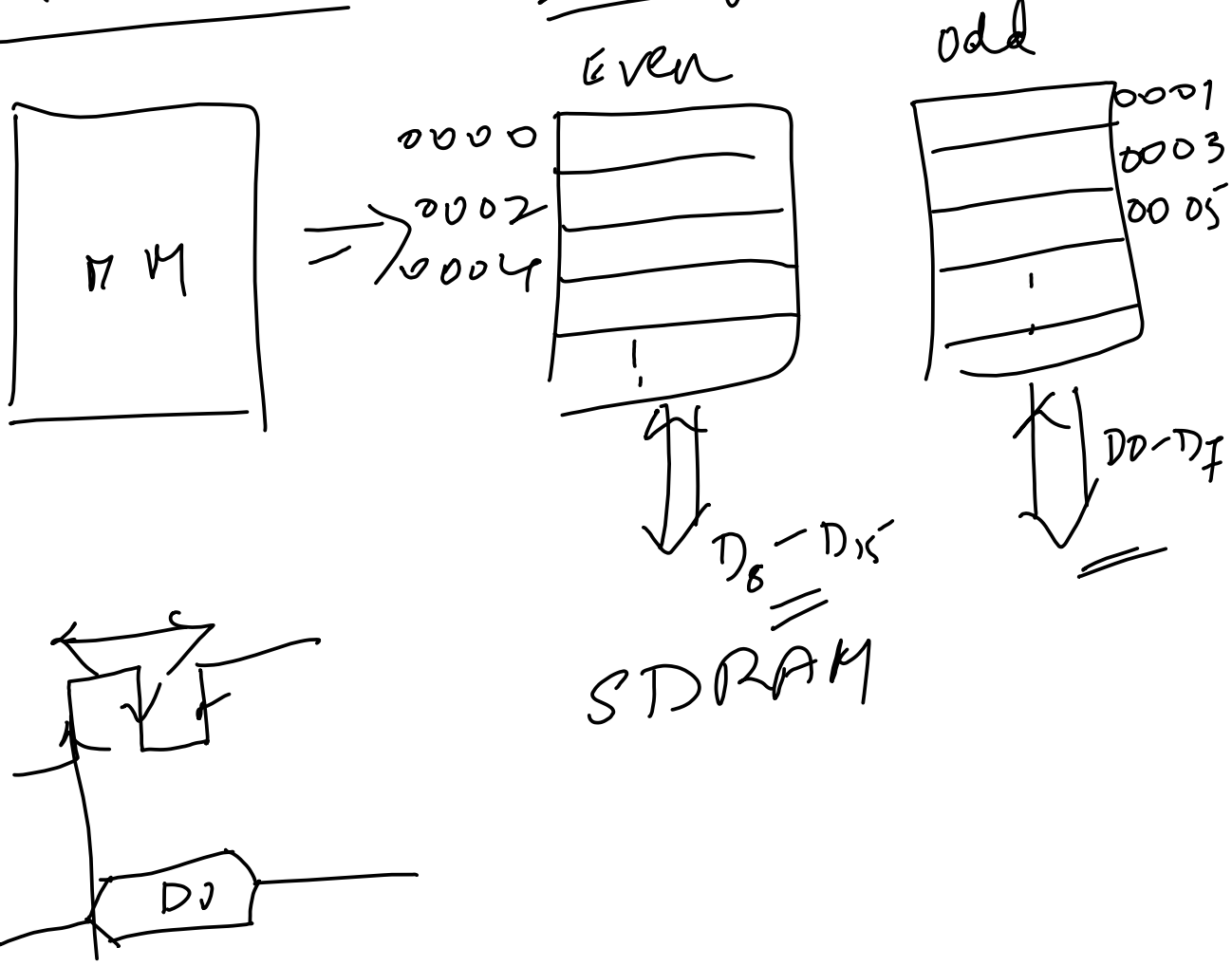
col

$A_0 - A_9$ $\Rightarrow$

$\Updownarrow$ $D_0 - D_{15}$

DDR - SDRAM

2-way - 2 banks



SRAM	DRAM
* requires constant flow of current to retain information	* requires constant refreshing to retain data
* less access time, faster	* slower because of increased access time
* cost per bit is high, expensive	* cost per bit is low, less cost

* less density
* more no. of transistors per mem cell

* more density
* one transistor per mem. cell
* SDRAM, Asynchronous DRAM

DRAM — Synchronous
— asynchronous

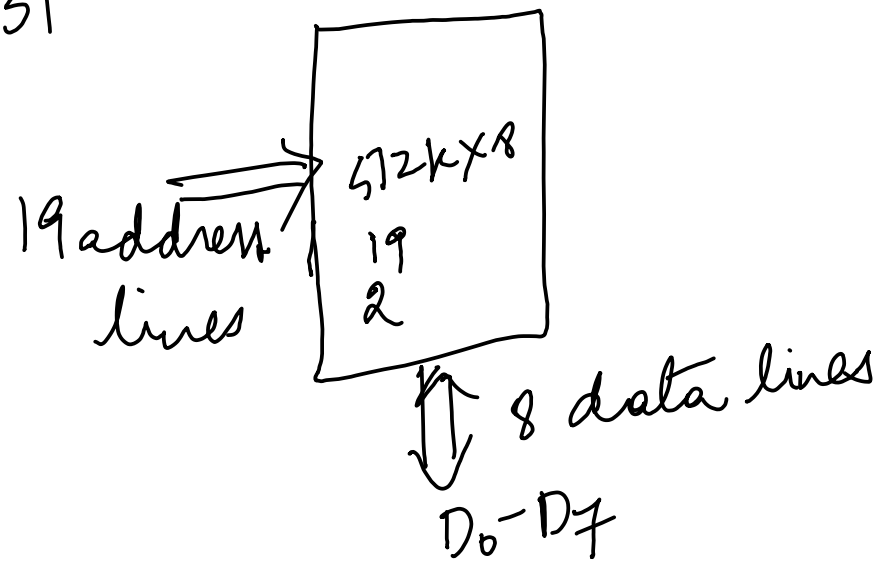
Synchronous DRAM	Asynchronous DRAM
*	

2M X 32 using 512K X 8

2^{21}

$D_0 - D31$

$A_0 - A_{20}$



2M X 32

$2^{21} \times 2^5$

$2^{19} \times 2^3$

512K X 8

$2^{19} \times 2^3$

$\Rightarrow 2^2 \times 2^2 = 4 \times 4 \Rightarrow$ 16 chips

$\Rightarrow$

2M X 16 using 256k x 8

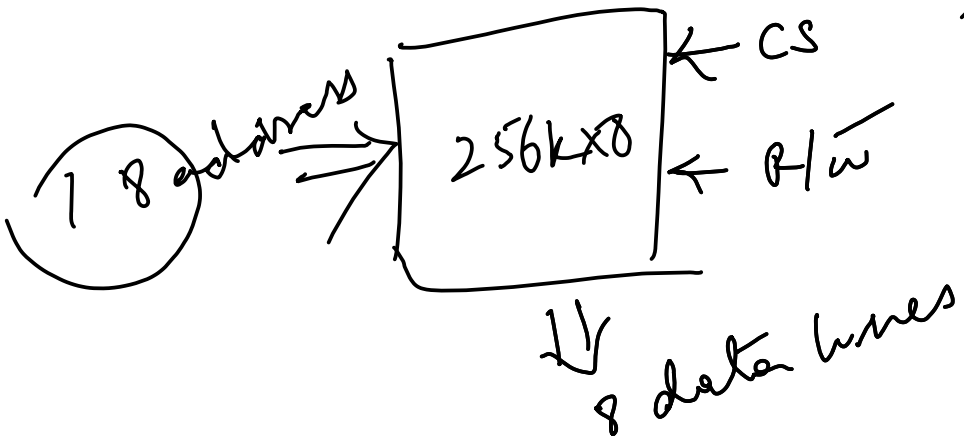
$$\frac{2M}{256k} \times \frac{16}{8} \Rightarrow \frac{2^{21}}{2^{18}} \times \frac{2^4}{2^3}$$

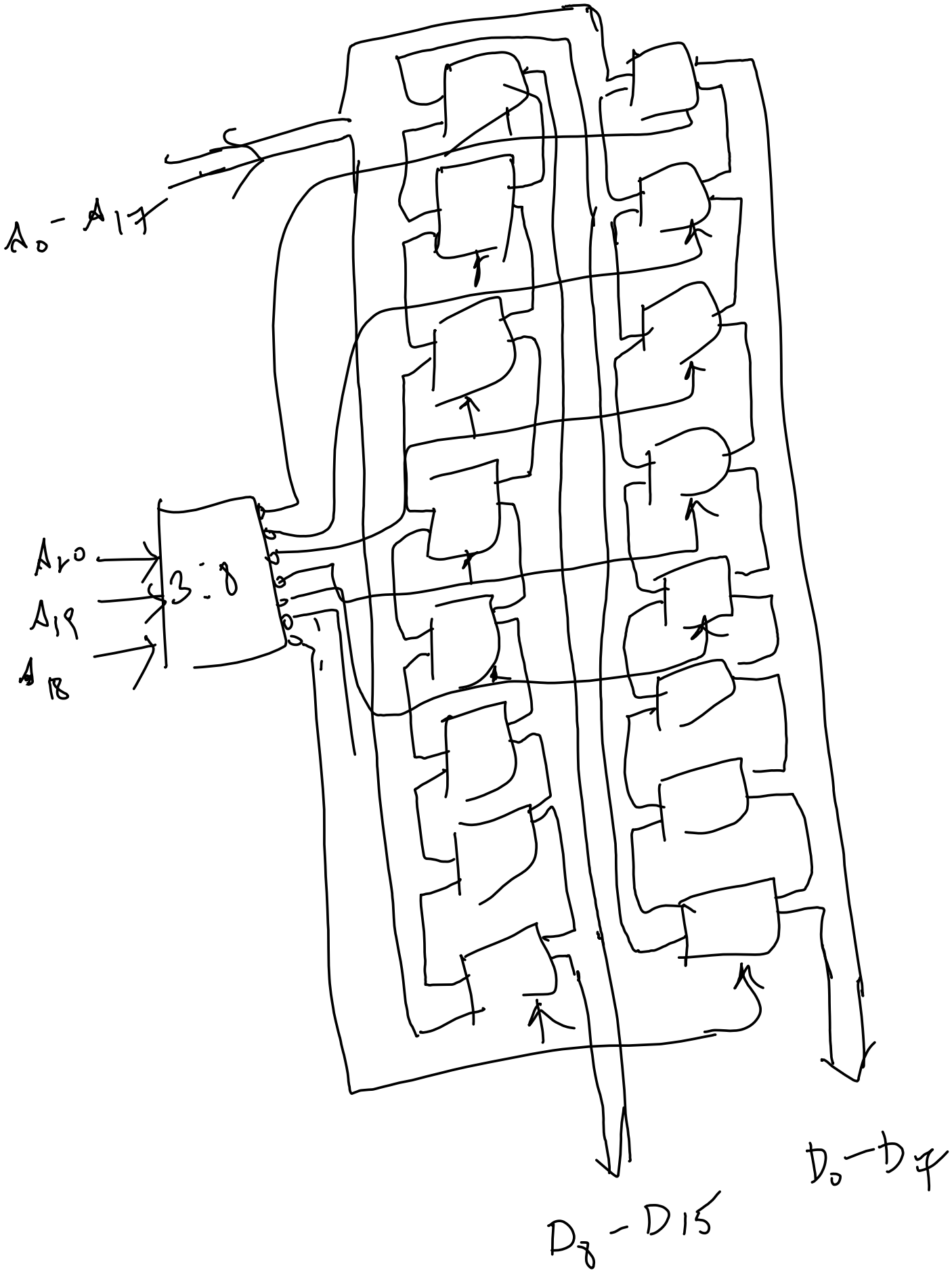
$$= 2^3 \times 2^1 = 8 \times 2 = 16$$

(3 → row, 1 → col)

Total address size = 21 bits ($A_0 - A_{20}$)

$21 - 18 = 3$




A0 - A17
A20
A19
A18
3:8
D0 - D7
D8 - D15